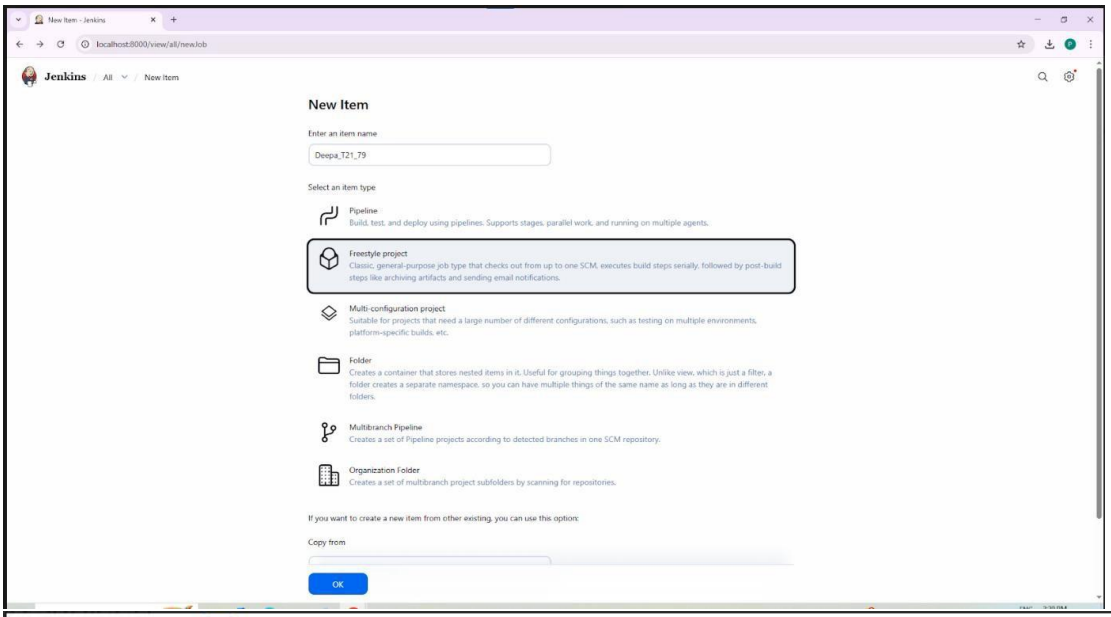
EXPERIMENT – 05

Aim: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Theory:

Continuous Integration and Continuous Deployment (CI/CD) is a software development practice that automates the process of building, testing, and deploying applications. Jenkins is a widely used open-source automation server that enables developers to create pipelines for integrating code changes, running tests, and deploying applications efficiently. In this experiment, build tools such as Apache Maven, Gradle, or Apache Ant are used to compile source code, manage dependencies, and package the application. A Jenkins pipeline is defines using a script (commonly called a Jenkinsfile), which specifies stages such as Build, Test, and Deploy. During the build stage, the selected tool (Maven/Gradle/Ant) compiles the project and generates an artifact (e.g., WAR file for web applications). In the testing stage, automated test cases are executed to ensure code quality. finally, in the deployment stage, the generated artifact is deployed to a web server such as Apache Tomcat, which hosts and runs Java-based web applications. This pipeline-based approach ensures faster development cycles, early detection of errors, and consistent deployment practices. By integrating build tools with Jenkins and deploying on Tomcat, organizations achieve automation, scalability, and reliability in their software delivery process.



3 Add Build Step

- Scroll to **Build**
- Click **Add build step**
- Select **Execute Windows batch command**

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4 Write command

bat

```
echo Hello Jenkins
```

- Click **Apply → Save**

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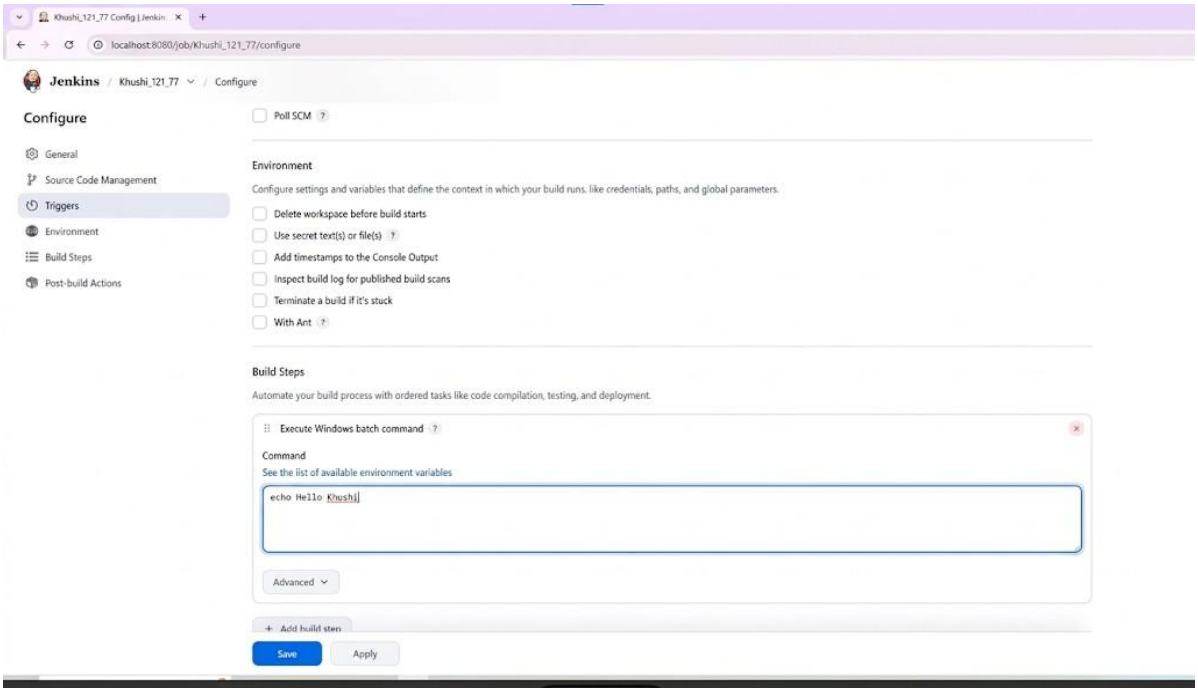
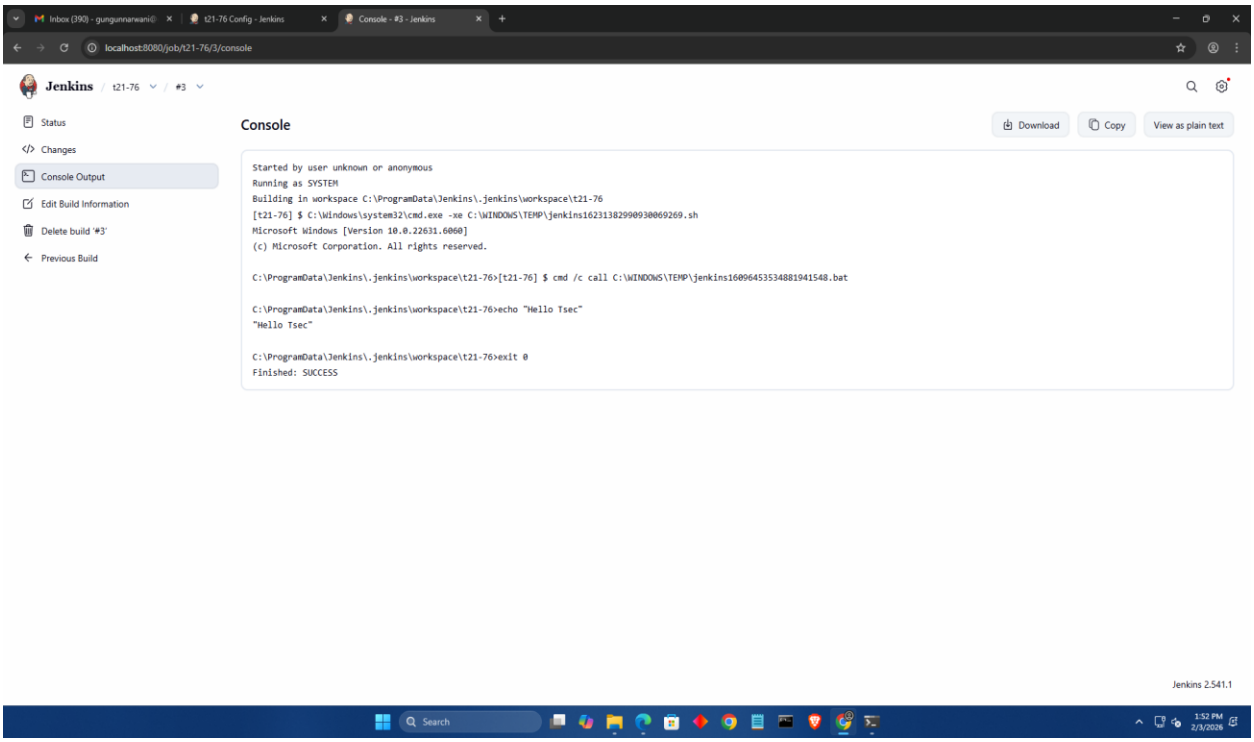
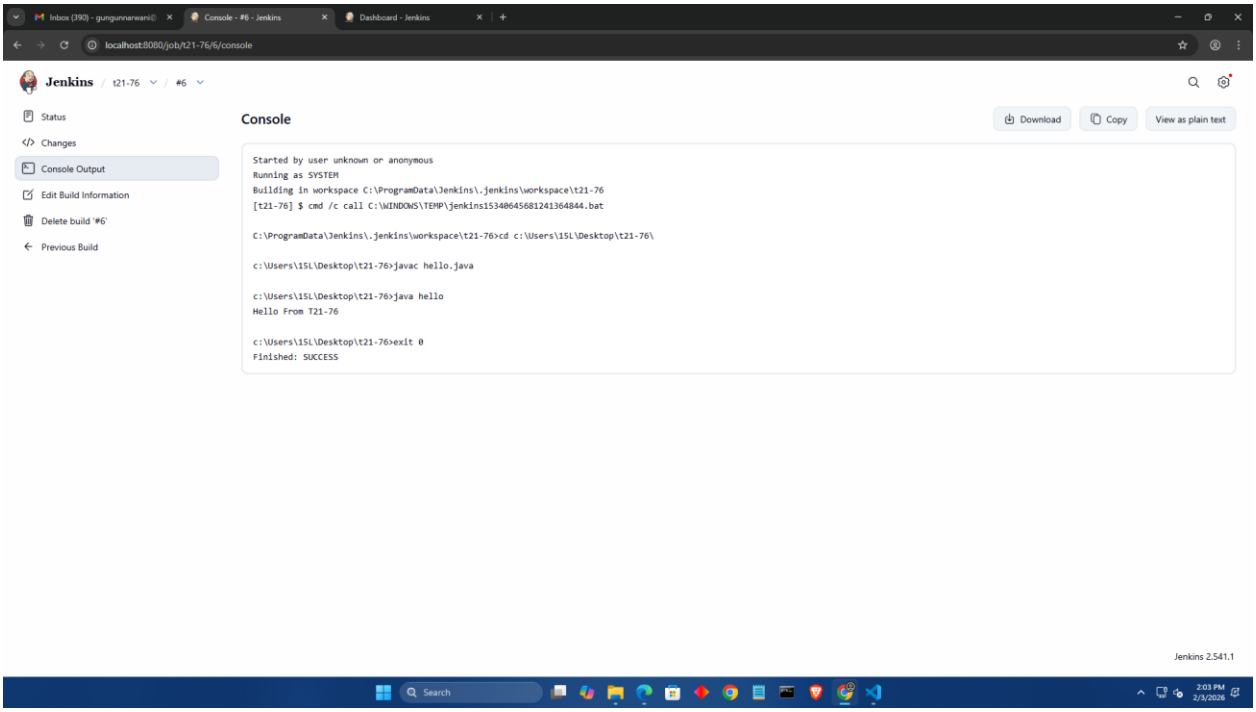
5 Build and Output

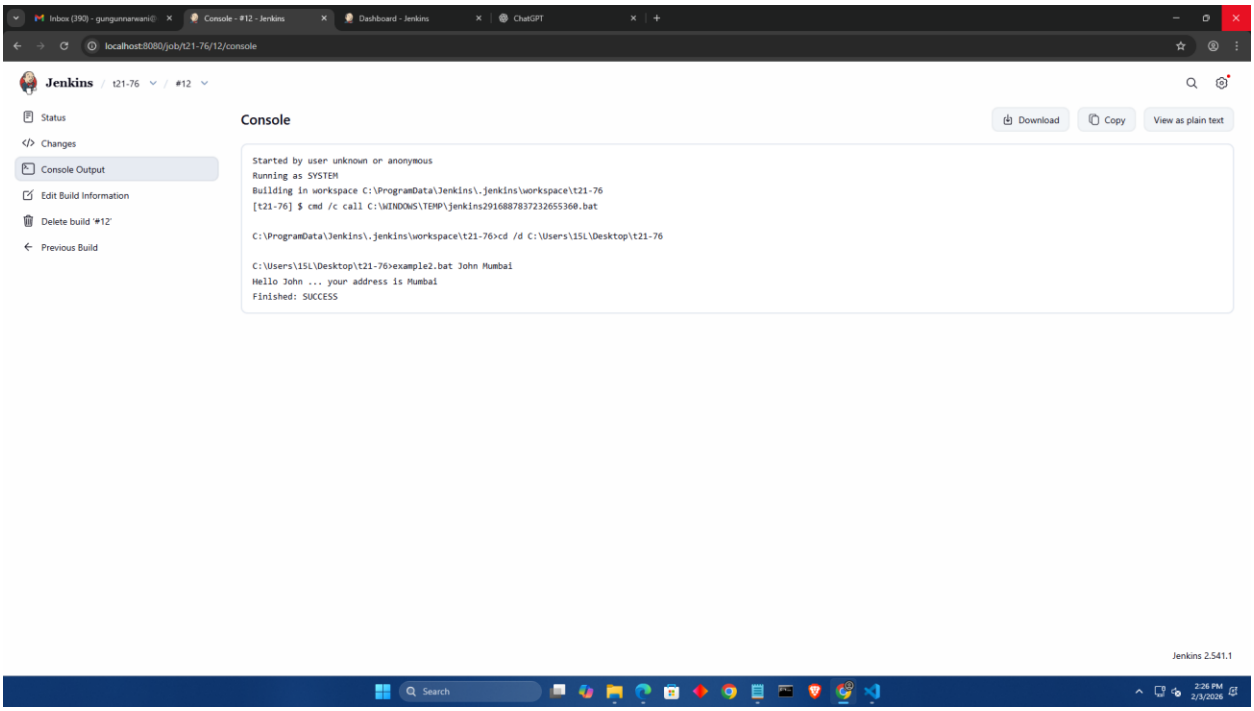
- Click **Build Now**


Khushi_121_77 Config [Jenkin
localhost:8080/job/Khushi_121_77/configure
Jenkins / Khushi_121_77 / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Poll SCM
Environment
Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.
Delete workspace before build starts
Use secret text(s) or file(s)
Add timestamps to the Console Output
Inspect build log for published build scans
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute Windows batch command
Command
See the list of available environment variables
echo Hello Khushi
Advanced
Add build step
Save
Apply

Gungun Narwani T21 Roll no:76







Conclusion: Successfully implemented the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server